

3rd Dec 2025

MIT MORADABAD

2nd CT

Course: B. Tech.

Department of Applied Sciences & Humanities

Semester: 1st

Session: 2025 – 26

Sections: E, F, G, H

Subject: Physics

Subject Code: BAS 101

MM: 20 Time: 1 Hr

Q. No.	1	2	3	4	5	6
CO / Blm	5 K2	2 K3	1 K3	5 K3	2 K4	1 K2

PART A

1	Write a short note on Quantum Dots.	2
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PART B

2	If the earth receives $2 \text{ cal min}^{-1} \text{ cm}^{-2}$ solar energy, what are the amplitude of electric and magnetic field of radiation?	3
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OR

If the upper atmospheric layer of the earth receives 1360 Wm^{-2} energy from the sun, what will be the peak values of electric and the magnetic fields at the layer?

3	An electron is bound in 1-D potential box which has width $2.8 \times 10^{-10} \text{ m}$. Assuming the height of box to be infinite, calculate the lowest 3 permitted energy values of electron.	3
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OR

X-ray of wavelength $0.015 \text{ \AA}$ is scattered at 60° . Find the wavelength of scattered X-ray.

PART C

4	Explain Type I and Type II superconductors? A superconducting tin has critical temperature of 3.7 K in zero magnetic fields and a critical field of 0.306 T at 0 K . Calculate the critical field at 2 K .	4
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OR

What is Meissner effect? Explain how Meissner effect proves the superconductor to be a perfect diamagnetic material? At what temperature is $H_c(T) = 0.1 H_c(0)$ for lead (Pb) having $T_c = 7.2 \text{ K}$.

5	Write down Maxwell's equations for free space and prove that the electromagnetic waves are transverse in nature.	4
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OR

Define Poynting Vector and explain its physical significance for electromagnetic wave in free space. Derive an expression for Poynting Theorem.

6	What do you understand physical significance of wave function. Derive Schrodinger Time independent and Time dependent wave equations.	4
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OR

Explain Compton Effect and derive an expression for the Compton shift.

- Write down proper theory wherever necessary.
- Draw Neat & Labeled Diagrams using Pencil only.

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20th Jun 2025

MIT MORADABAD

2nd CT

Department of Applied Sciences & Humanities

Course: B. Tech.

Semester: 2nd

Session: 2024 – 25

Sections: A,B,C,D

Subject: Physics

Subject Code: BAS 201

MM: 20

Time: 1 Hr

Q. No.	1	2	3	4	5	6
CO/ Blm	5 K2	2 K3	1 K3	5 K3	2 K4	1 K2

PART A

1.	Write a short note on Quantum Dots.	2
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PART B

2.	If the earth receives $3 \text{ cal min}^{-1} \text{ cm}^{-2}$ solar energy, what are the amplitude of electric and magnetic field of radiation?	3
3.	An electron is bound in 1-D potential box which has width $2.8 \times 10^{-10} \text{ m}$. Assuming the height of box to be infinite, calculate the lowest 3 permitted energy values of electron.	3

PART C

4.	Explain Type I and Type II superconductors? A superconducting tin has critical temperature of 3.7K in zero magnetic fields and a critical field of 0.306 T at 0 K. Calculate the critical field at 2 K.	4
5.	Write down Maxwell's equations for free space and prove that the electromagnetic waves are transverse in nature.	4
6.	What do you understand physical significance of wave function. Derive Schroedinger Time independent and Time dependent wave equations. OR Explain Compton Effect and derive an expression for the Compton shift. X-ray of wavelength $0.015 \text{ \AA}$ is scattered at 60° . Find the wavelength of scattered X-ray.	4

- Write down proper theory wherever necessary.
- Draw Neat & Labeled Diagrams using Pencil only.

18.06.2025